SOFTWARE REQUIREMENTS IN SCRUM

<epam> | Campus

October 27, 2025

AGENDA

01. Team introduction
02. Who is Business Analyst?
03. Software Development Methodologies
04. Product Requirements
05. MVP
06. User story
07. Home task

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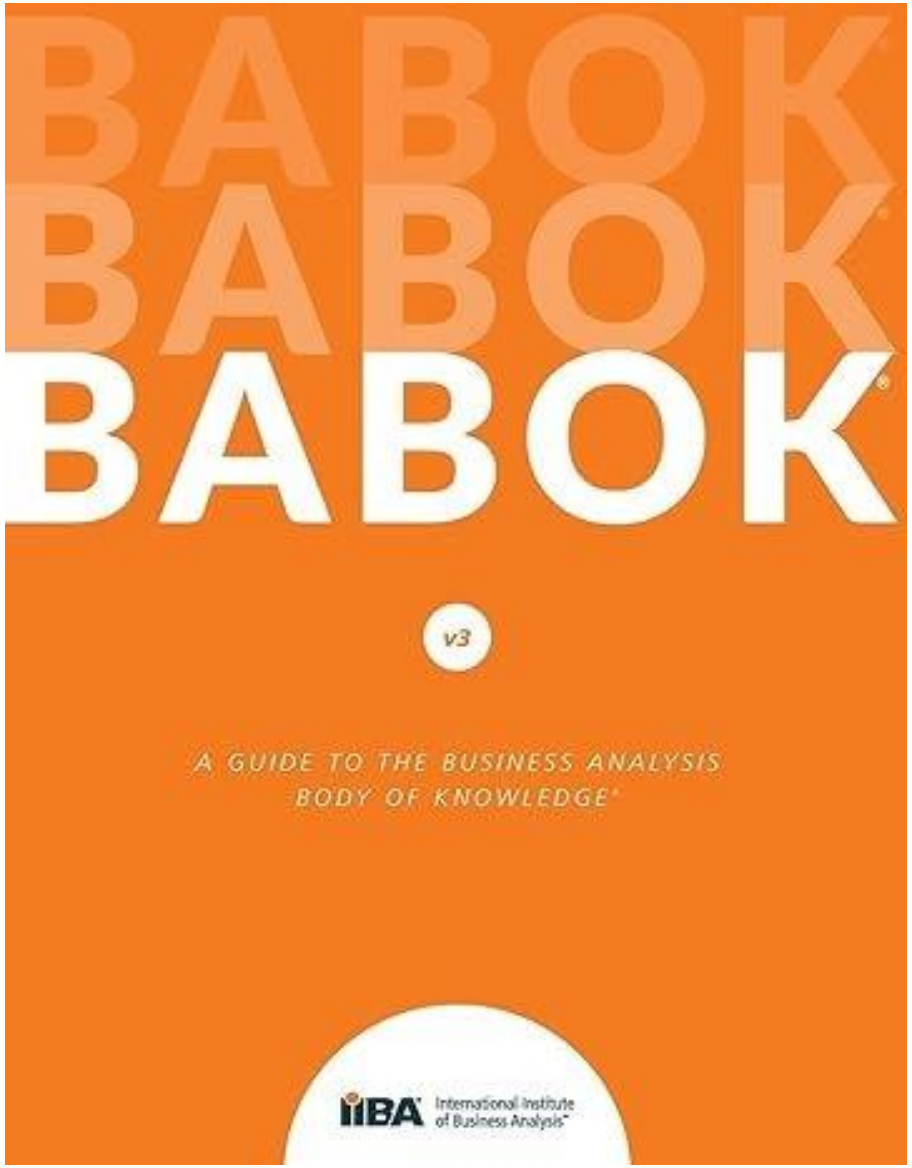
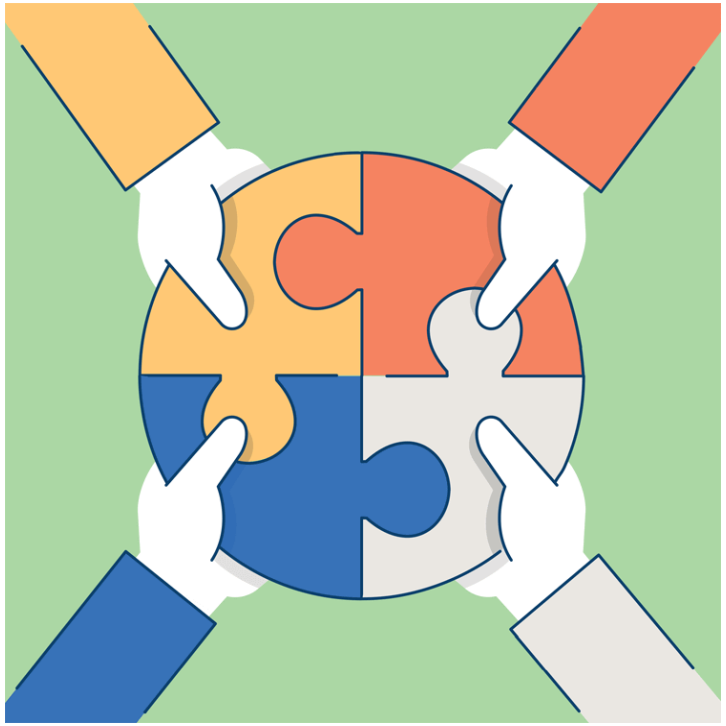
Business Analysis – what and why?

Business Analysis is the practice of enabling change in an organizational context, by defining needs and recommending solutions that deliver value to stakeholders.



Stakeholders?

Someone with an interest in the outcome of a project, either because they have funded it, will use it, or will be affected by it.

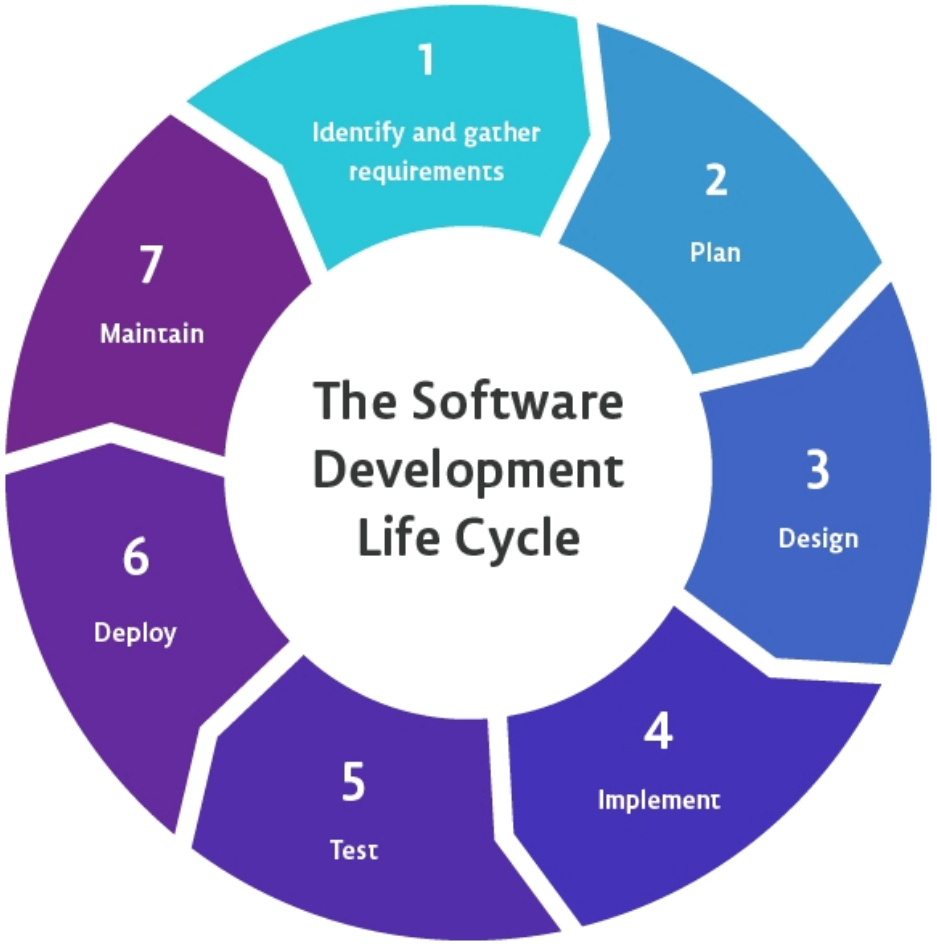


A group or person who has interests that may be affected by an initiative or have influence over it.

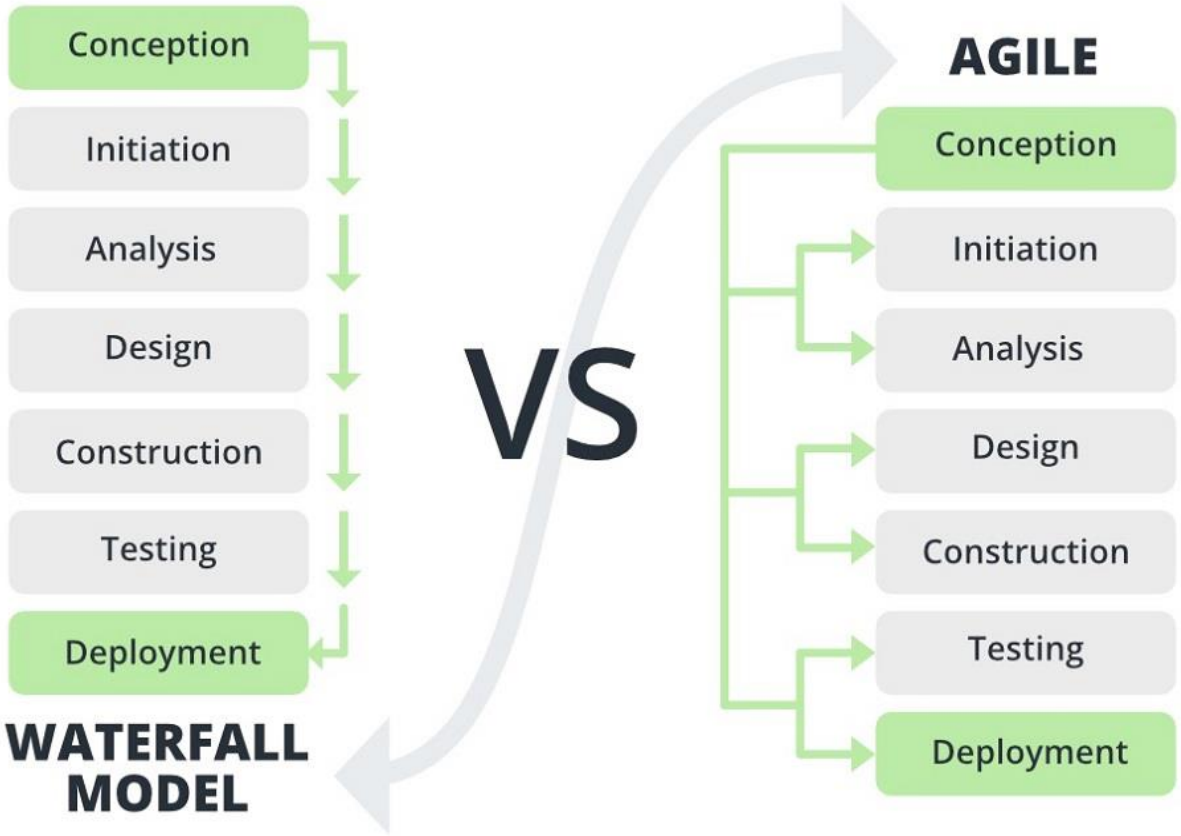
Business Analyst – who are you?

A **Business Analyst** is any person who performs business analysis tasks described in the BABOK®Guide, no matter their job title or organizational role.

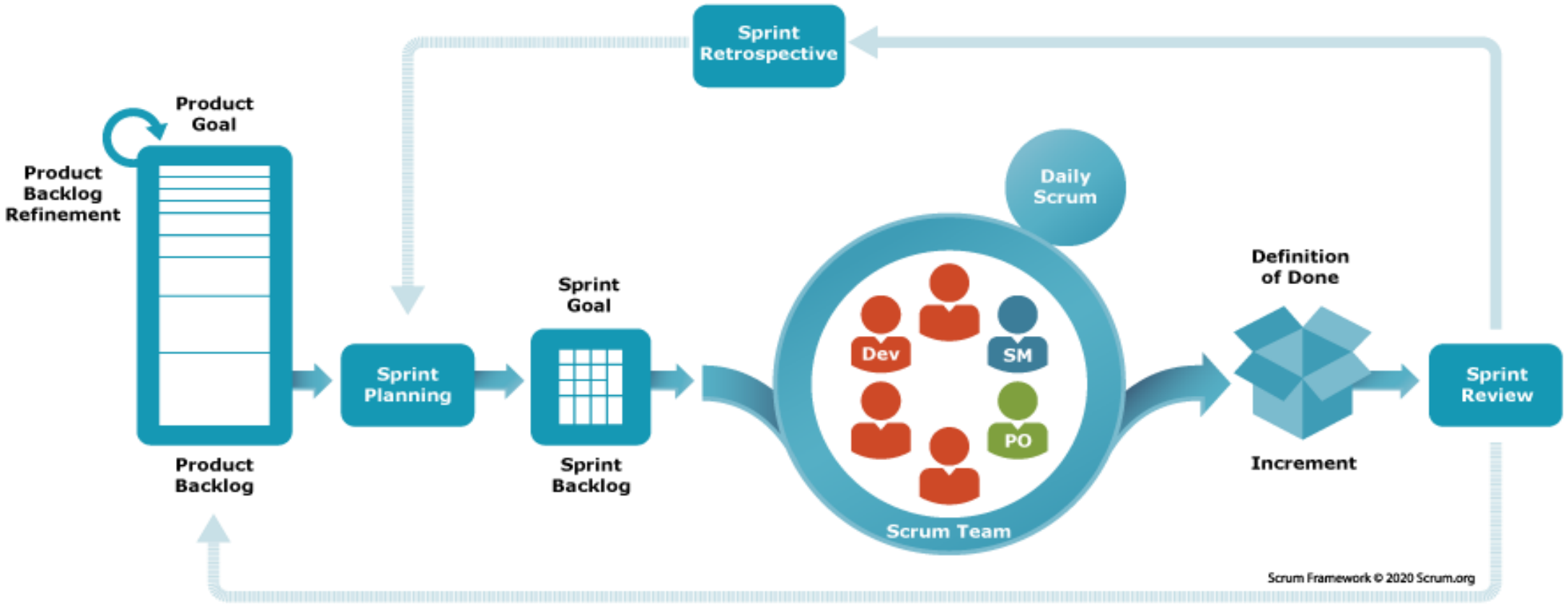
Software Development Lifecycle



Software Development Methodologies



Scrum Overview

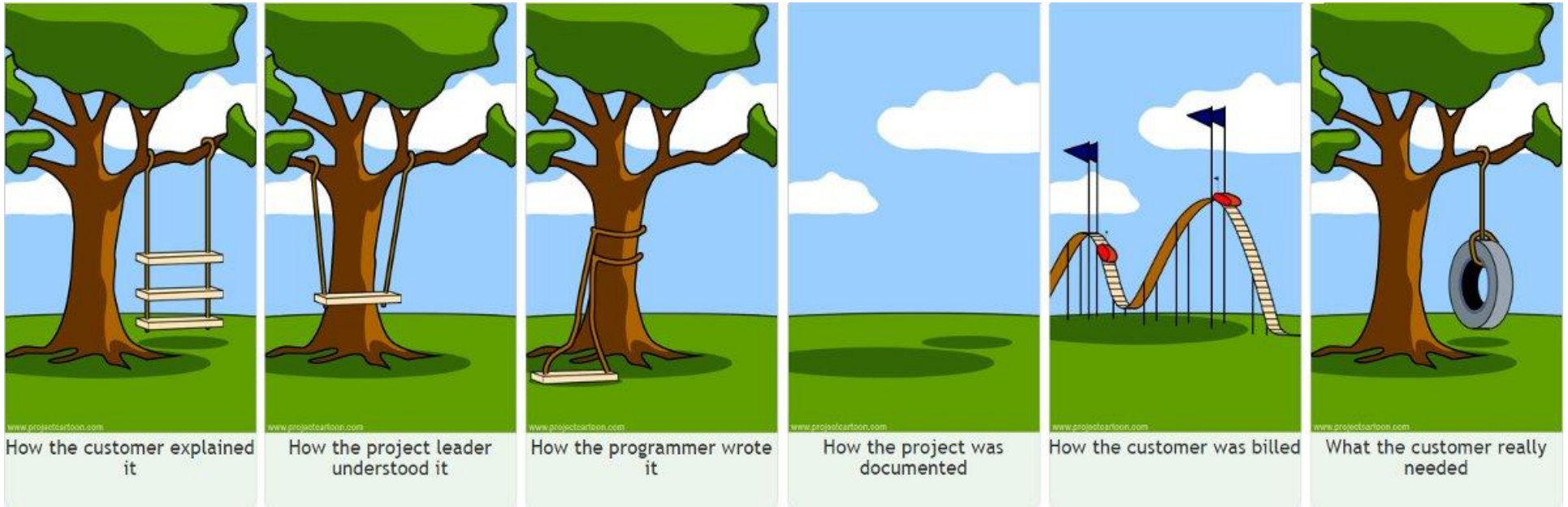


Scrum in Action

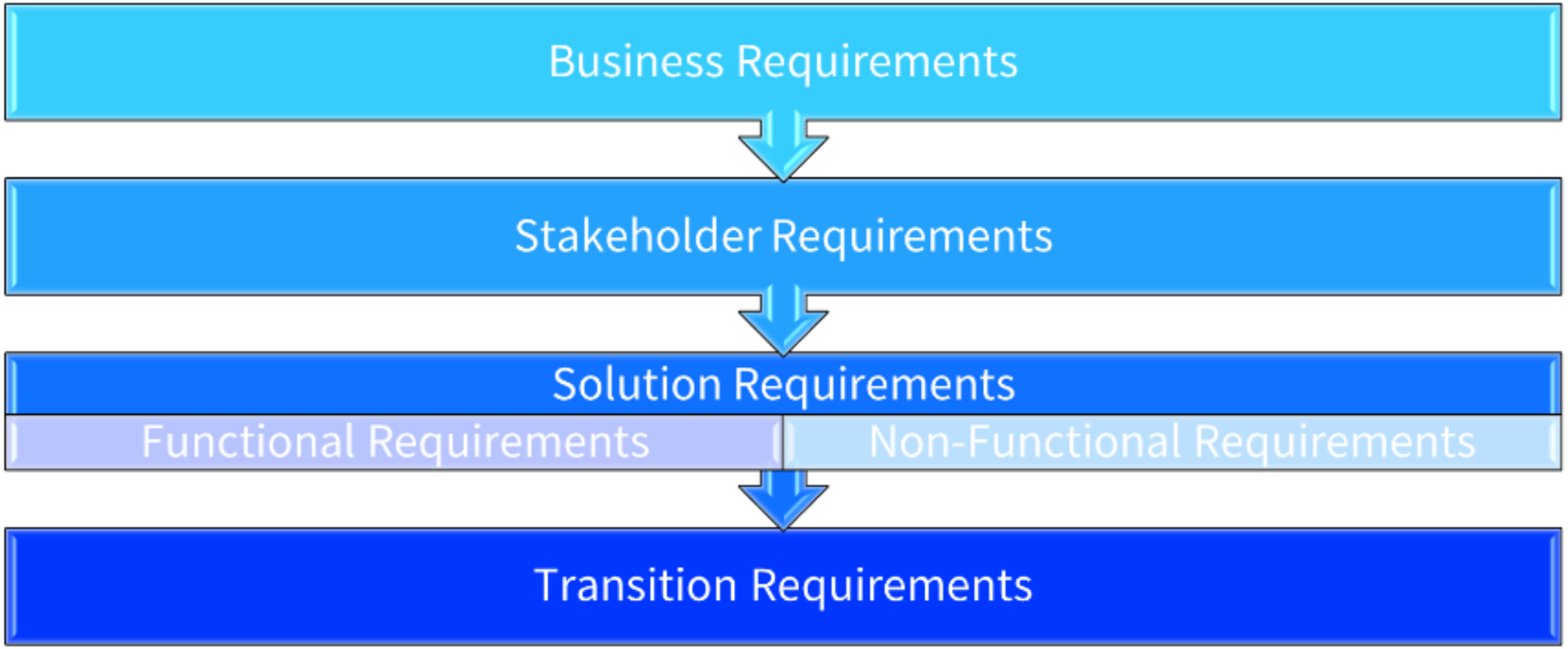
1. Sprint planning before the start of each sprint
2. Start the sprint: bi-weekly on Friday
3. Daily Scrum meetings 3 times a week: on Monday, Wednesday and Friday, 15-30 minutes
4. Working increment till the end of the sprint
5. Sprint review on the end date of the sprint to demo the working increment

Requirements – what is it?

- A condition or capability needed by a stakeholder to solve a problem or achieve an objective.
- A condition or capability that must be met or possessed by a solution or solution component to satisfy a contract, standard, specification, or other formally imposed documents.
- A documented representation of a condition or capability as in (1) or (2).



What types of requirements are there?



Characteristics of "good" requirements

Specific – a requirement should express exactly what is required. It means it is clear and unambiguous.

Measurable – verification of the fact that a requirement has been implemented in the right way should be possible.

Attainable – implementation of a requirement within the given context should be possible.

Relevant – a requirement should be relevant to the needs of a specific organization.

Traceable – requirements should be linked to relevant designs and test cases.

Business Goals for our product

Key Business Goals

- Demonstrate the full SDLC in an educational and engaging way
- Ensure the product is reproducible by participants, regardless of their technical background
- To provide an opportunity for students to finalize the project, materials from which they will be able to use for their diploma
- Provide ability for students to prepare the working project which they could add to own portfolio

Product intro

What will we do?

Secret Nick

Application designed to facilitate and streamline the organization of gift exchange events

Functional Requirement Example

REQ – 01 Room creation

Description:

The system must allow user to create a new Secret Nick room through a structured three-step process.

Acceptance criteria:

- The room must be created through a structured three-step process including the following:
  - Step 1: Room Data Input
  - Step: User data input
  - Step 3: User's wishlist creation
- Upon the room created:
  - The user becomes the room's Administrator
  - The user is automatically included as a participant in the game

Functional Requirement Example

REQ – 05 Randomizer start

Description:

The system must allow Administrator to start the randomization process at any time after the minimum number of participants (3) is reached. Each participant is randomly assigned a gift recipient.

Acceptance criteria:

- The randomization process can only be initiated once per room
- The randomization process can be initiated by Administrator (Room creator) at any time
- The randomizer assigns each participant a unique recipient
- The system must verify that the participant count is at least 3 before allowing randomization to start
- After the randomization has been initiated, the randomization functionality becomes disabled for the Administrator

Functional Requirement Example

REQ – 10 Participant removal by Administrator

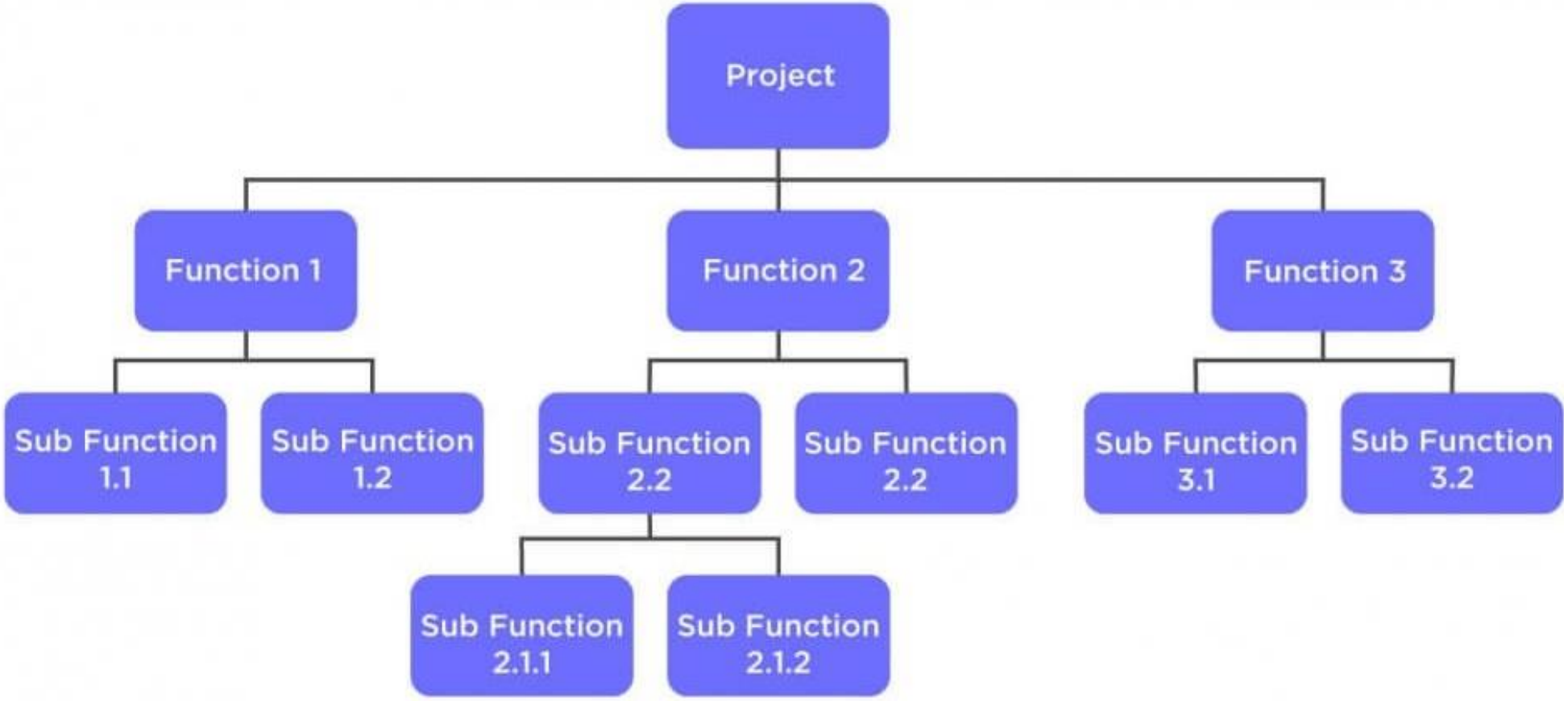
Description:

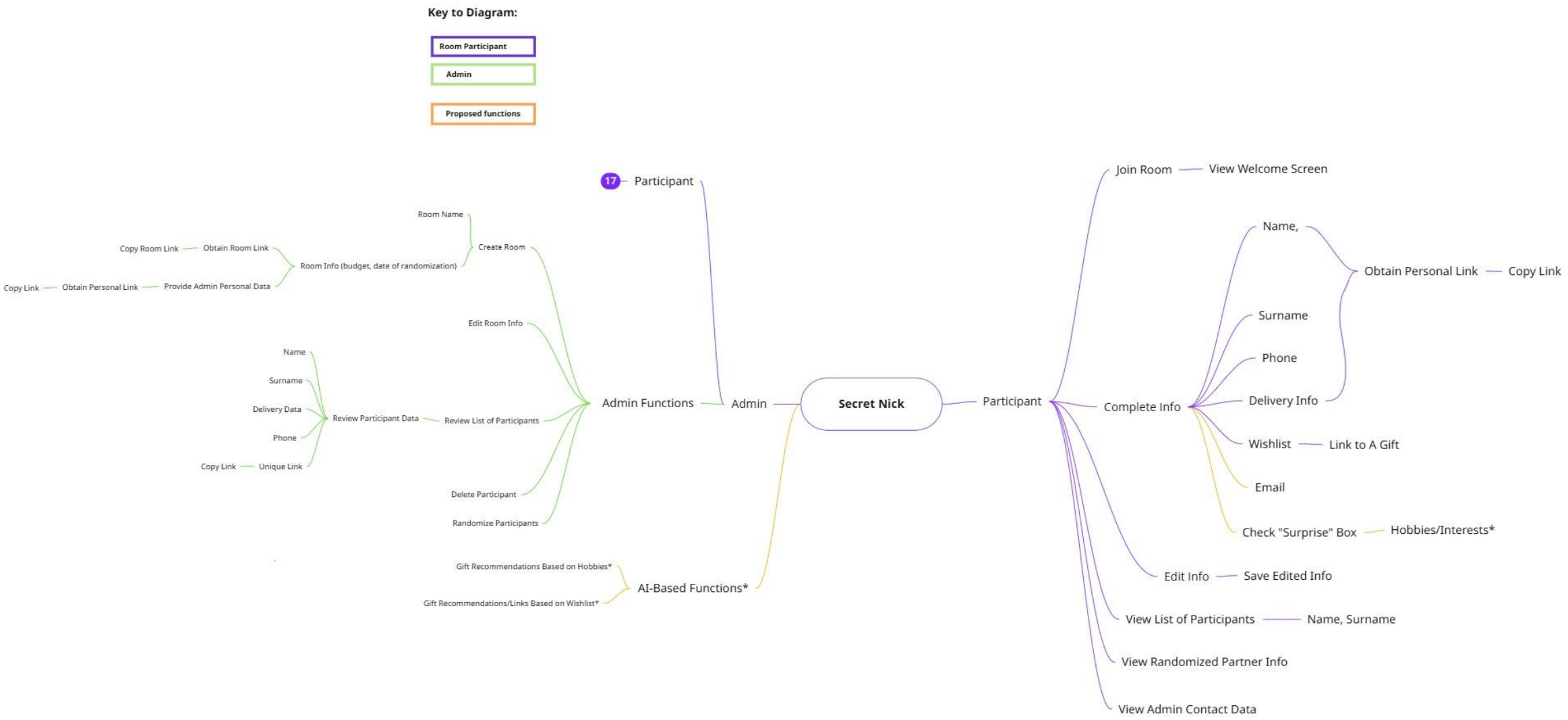
The system must allow the Administrator to remove participant from the room until the randomization starts

Acceptance criteria:

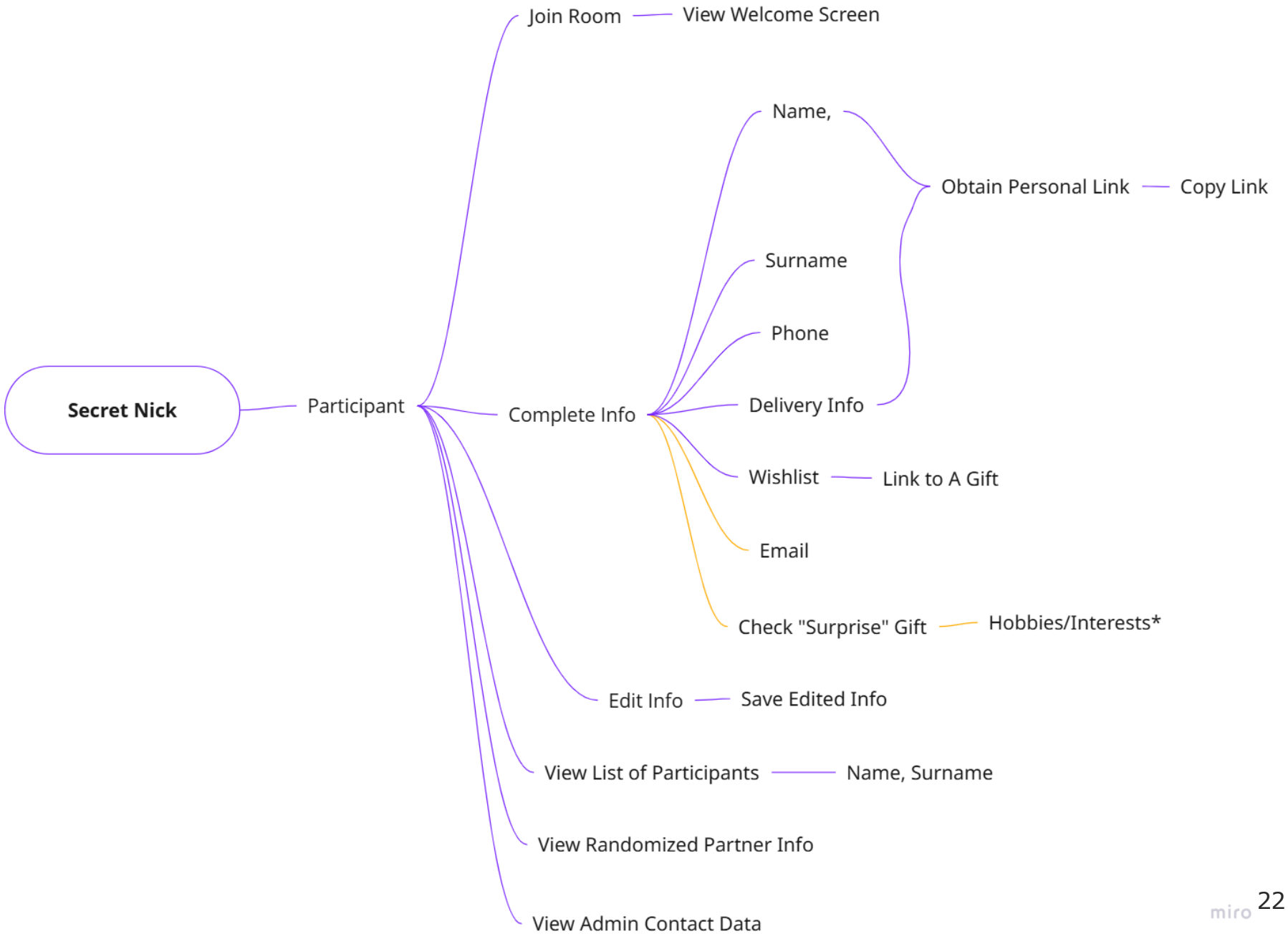
- The Administrator can remove any participant from the room until the randomization starts
- Removed participant lose access to the room and their data is deleted
- The system updates the number of participant in the room accordingly

Functional Decomposition

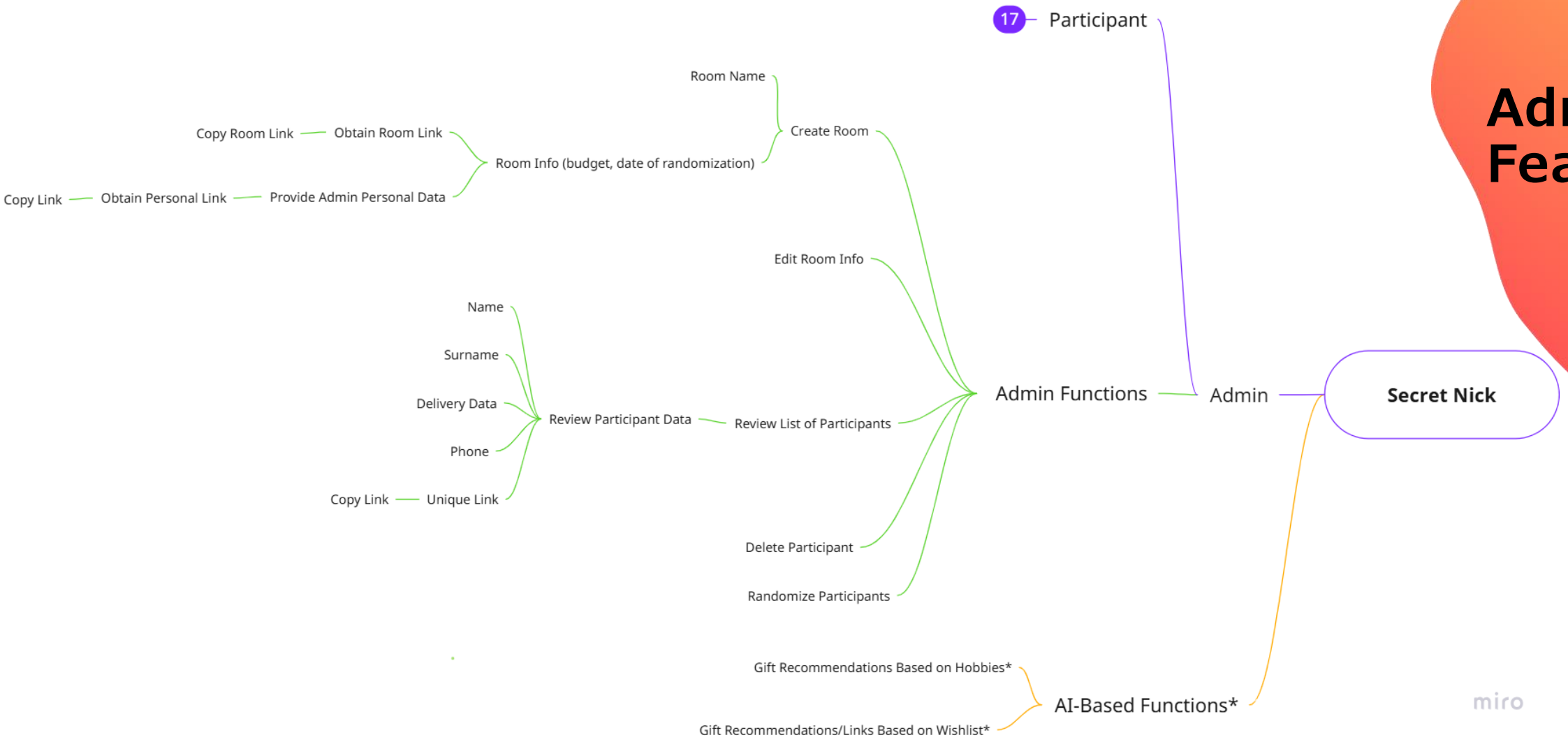



Key to Diagram:
Room Participant
Admin
Proposed functions
Secret Nick
Participant
Join Room
View Welcome Screen
Complete Info
Name,
Surname
Phone
Delivery Info
Obtain Personal Link
Copy Link
Wishlist
Link to A Gift
Email
Check "Surprise" Box
Hobbies/Interests*
Edit Info
Save Edited Info
View List of Participants
Name, Surname
View Randomized Partner Info
View Admin Contact Data
Admin
17
Participant
Admin Functions
Create Room
Room Name
Room Info (budget, date of randomization)
Obtain Room Link
Copy Room Link
Provide Admin Personal Data
Obtain Personal Link
Copy Link
Edit Room Info
Review List of Participants
Review Participant Data
Name
Surname
Delivery Data
Phone
Unique Link
Copy Link
Delete Participant
Randomize Participants
AI-Based Functions*
Gift Recommendations Based on Hobbies*
Gift Recommendations/Links Based on Wishlist*

Participant`s Features



Admin`s Features



miro

Prioritization Techniques of an MVP

M — **MUST HAVE**

Project cannot do without them

- ❑ Create a Room
- ❑ Randomize Participants
- ❑ View the randomized partner

S — **SHOULD HAVE**

Required over the long run

- ❑ View Participant list
- ❑ Admin review of Participant's data
- ❑ Editing functions

C — **COULD HAVE**

Low-cost tweaking

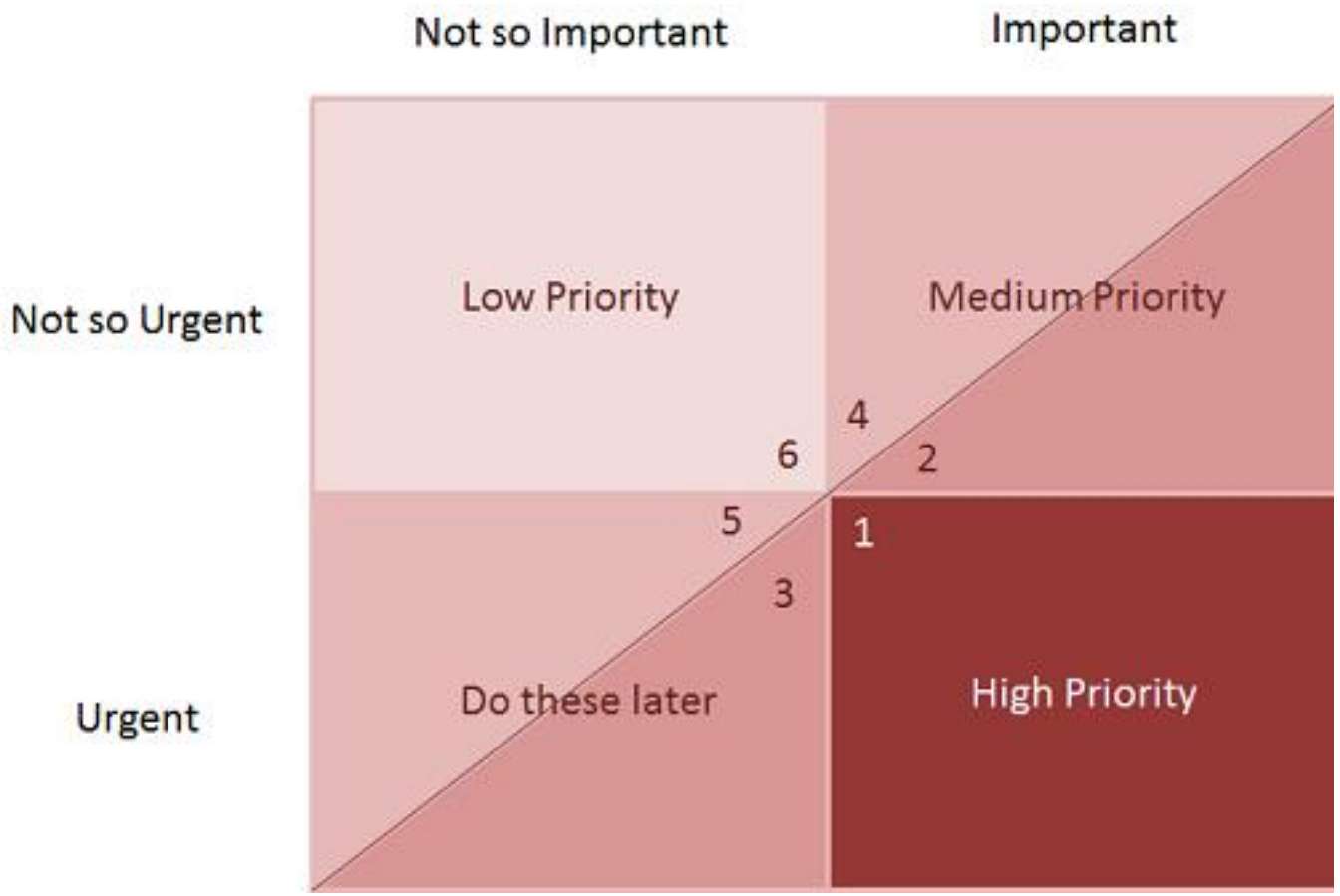
- ❑ Automatic start of randomizer
- ❑ Remove Participant from the room
- ❑ Set room expiration time (e.g. 30 days)

W — **WON'T HAVE**

Get back to them at better days

- ❑ Provide waybill number for the gifts
- ❑ Join the room with a unique code

Prioritization: Ranking

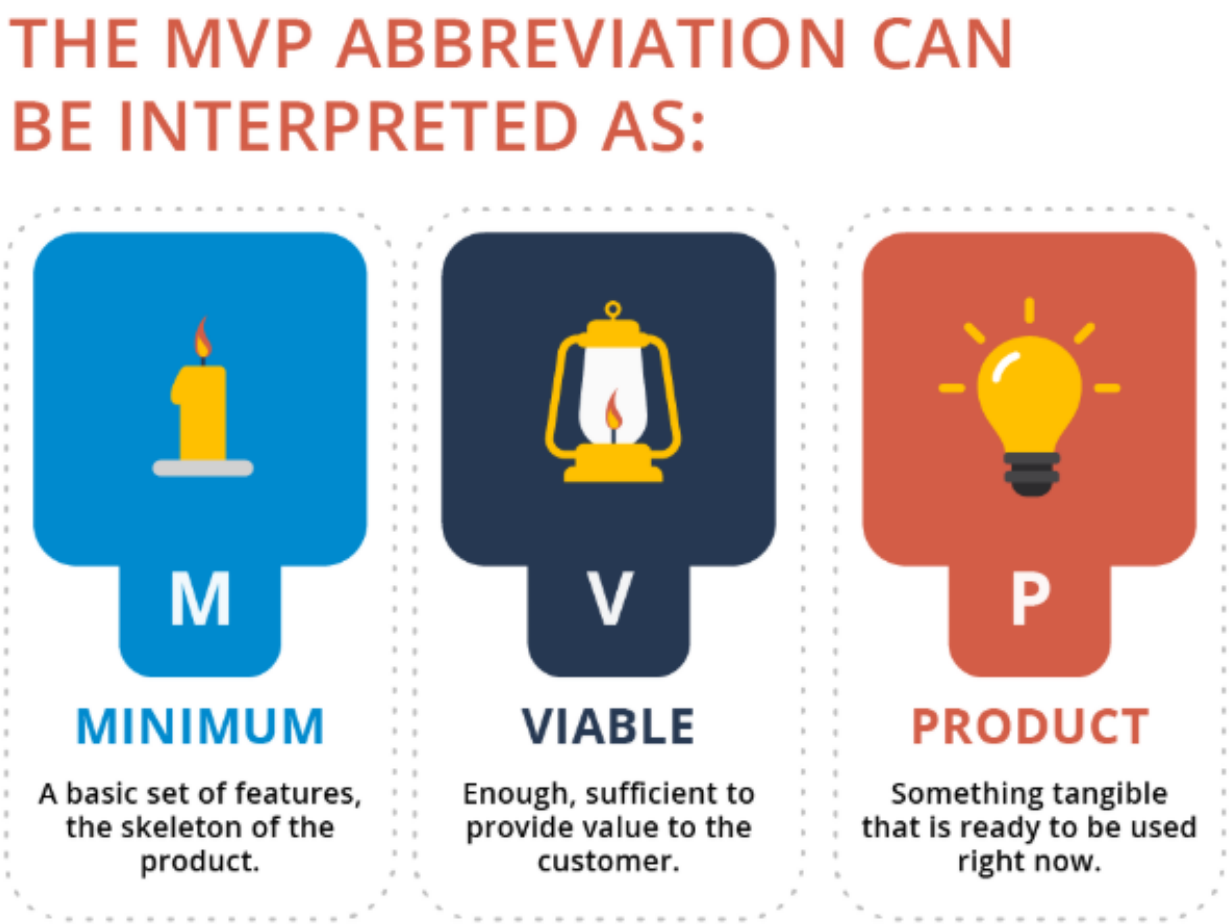


Important & Urgent:
- Create Room & Welcome pages
- Join the Room
- Randomize Participants
- View Giftee's data

Important & Not So Urgent:
- Editing room data & Wishlist
- Admin's review of Participant's data (phone, delivery info)
- Data validation

Urgent & Not So Important:
- Display room name on Success pages
- Favicon

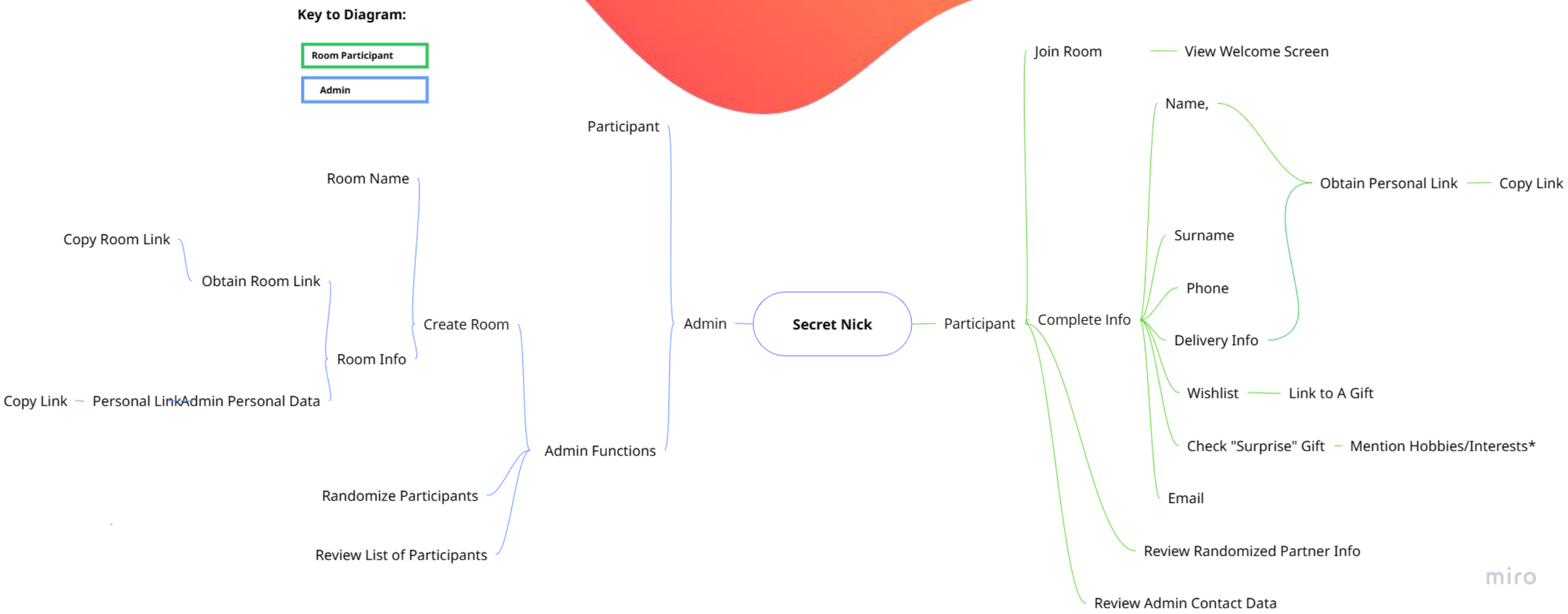
Not So Urgent & Not So Important:
- Remove Participants
- Participant info section



Purpose of a Minimum Viable Product

- To Enter the Market with a Small Budget
- To Collect the Maximum Quality Feedback
- To Find the Right Audience
- To Minimize Errors

Features to be in MVP



How requirements are documented in Scrum?

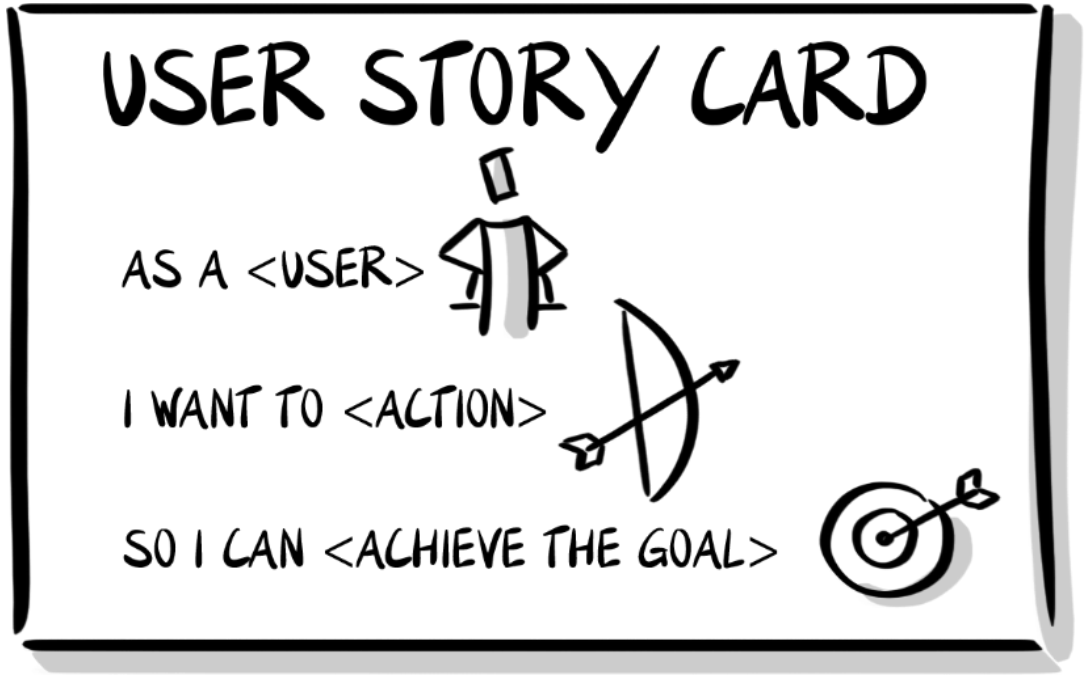
In Scrum, Requirements are Articulated as User Stories within the Product Backlog

Product Backlog is an ordered list of everything that might be needed in the product, providing a single and accountable source of requirements in Scrum.

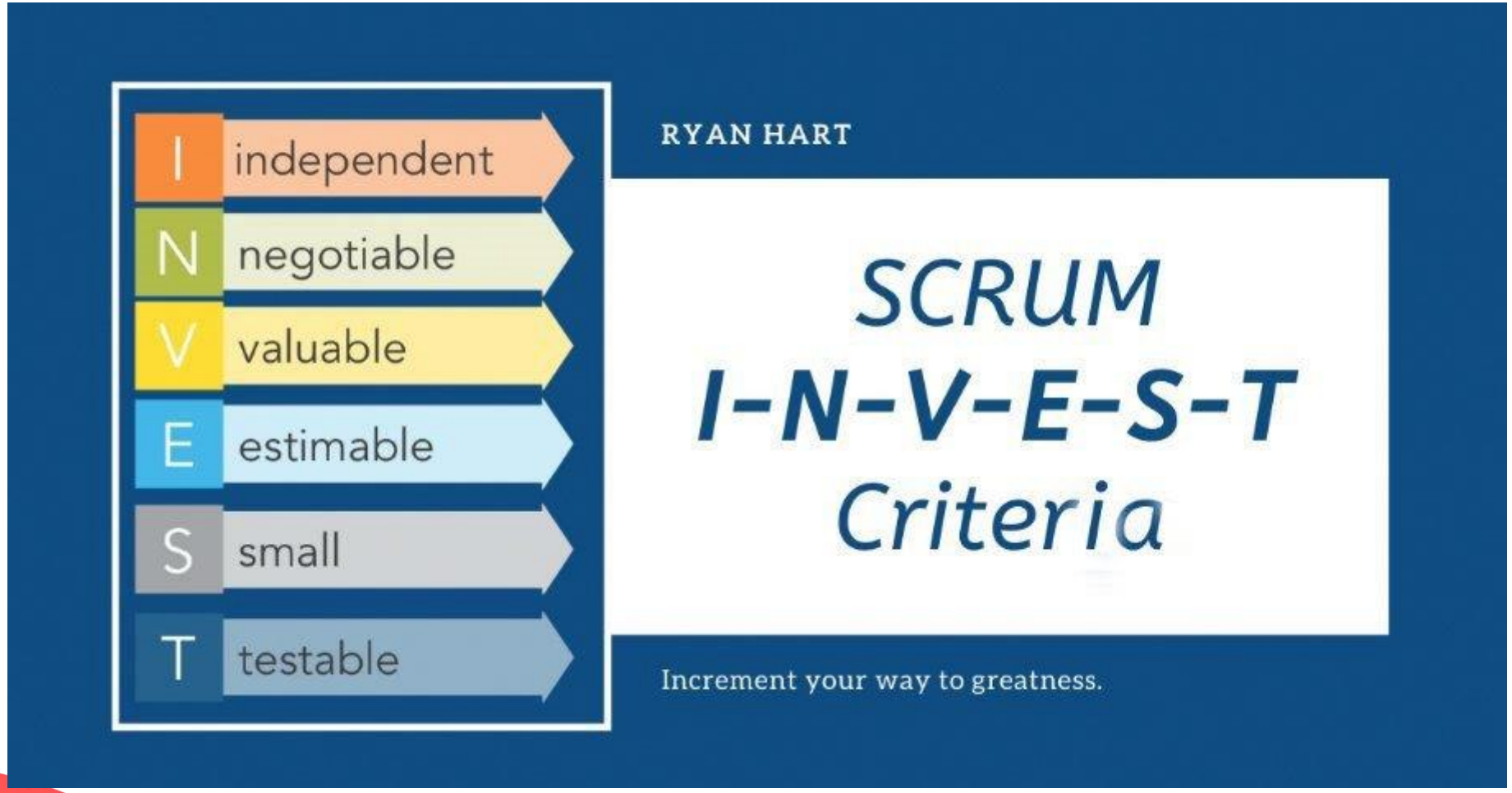
REQ-01 | Room Creation	MVP
REQ-02 | Room and Personal Link Generation after room creation	MVP
Welcome Page | Join the Room	MVP
REQ-04 | Participant Data Input	MVP
3 REQ-04 | Participant Personal Link generation	MVP
5 View Room Info Section	MVP
REQ-08 | Display of Participants List	MVP
REQ-11 | Randomization	MVP

User story – what it is?

Short, simple description of a feature from the user perspective, focusing on the value delivered to the user and fostering better communication.



What is a "good" user story?



User story Example

User story name:

REQ-11 | Randomization start

As an Admin,

I want to initiate the randomization process

for assigning Secret Nick partners,

So that participants are uniquely matched to another for gift exchange.

AC1: Randomizing Conditions

Given the Admin is in the room,
When the minimum number of Participants is reached (>=3)
Then the system enables the "start randomization" action for the Admin

AC2: Randomization Logic

Given the Admin has started the randomization,
When the randomization process run,
Then the system:

- Randomly assigns each participant to another participant in the same room,
- Ensure that no participant is assigned to themselves,
- Ensure that each participant is assigned exactly one recipient, and each participant is the recipient of exactly one other participant (one-to-one mapping),
- Guarantees no duplicate or missing assignments.

AC 3: **View assigned recipient details after randomization**

Given the randomization process is completed,
When a participant accesses the room,
Then the system displays:

- By default: assigned recipient's first and last name.
- Upon "Read details" action: recipient's phone, email, delivery address, and wishlist/chosen Surprise Gift option.

AC4: **Insufficient Number of Participants**

Given the Admin is in the room,

When the number of Participants in the room is fewer than 3 (<3), **and** an Admin is trying to start a randomizer,

Then the system displays a message: " *You need at least 3 people in the room to enable drawing.*", **and** the randomizing does not start

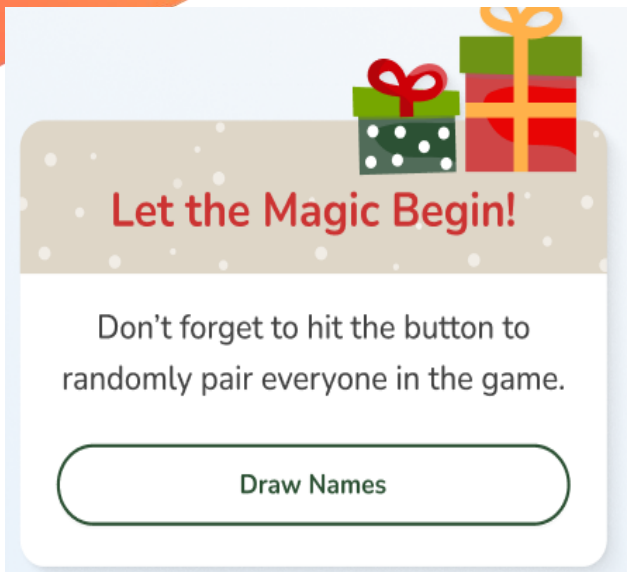
AC5: **Unsuccessful Request**

Given the Admin starts the randomization,

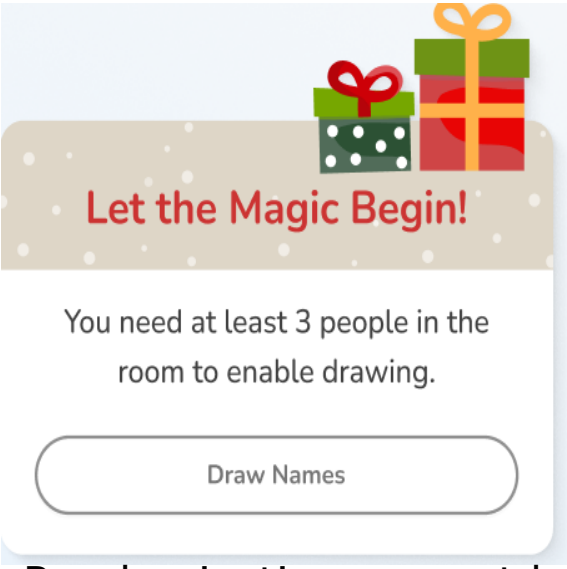
When the backend fails to respond,

Then the system displays a message: "*Something went wrong. Try again*".

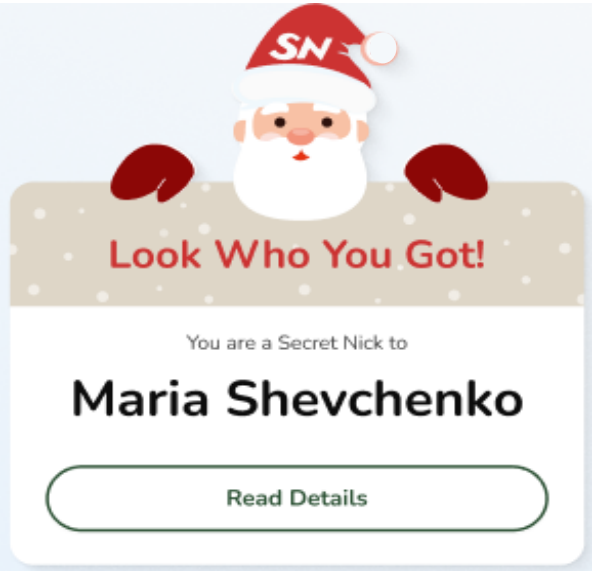
Are all AC mentioned in user story met?



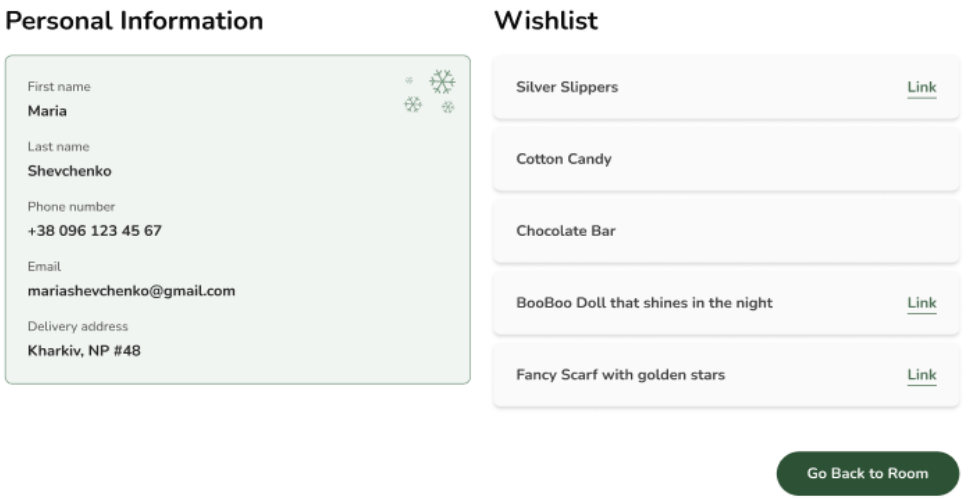
Randomization enabled



Randomization can not be started until the number of Participant in the room < 3

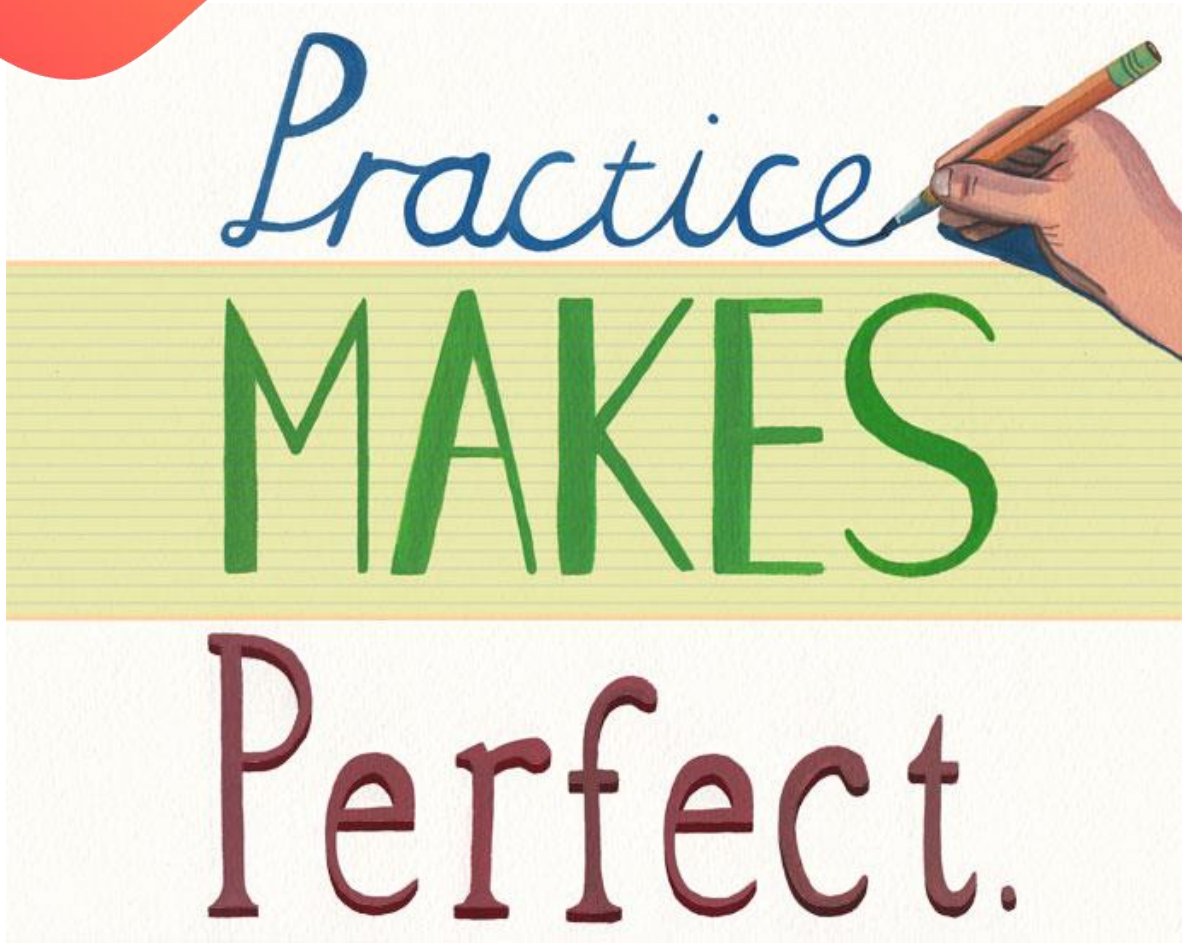


Assigned recipient after randomization



Recipient's details

Home task



User story preparation

- Prepare the user story for the functionality of a participant removal by Administrator (check the functional requirement given on a slide 19)
- User story must include user story card and acceptance criteria
- Feel free to propose the details of the user flow on your own and make sure you mention it in the acceptance criteria

How are you?

Questions?